Ex : 2
Date : 1/2/26

DETERMINATION OF CURRENT IN CIRCUIT USING MESH ANALYSIS

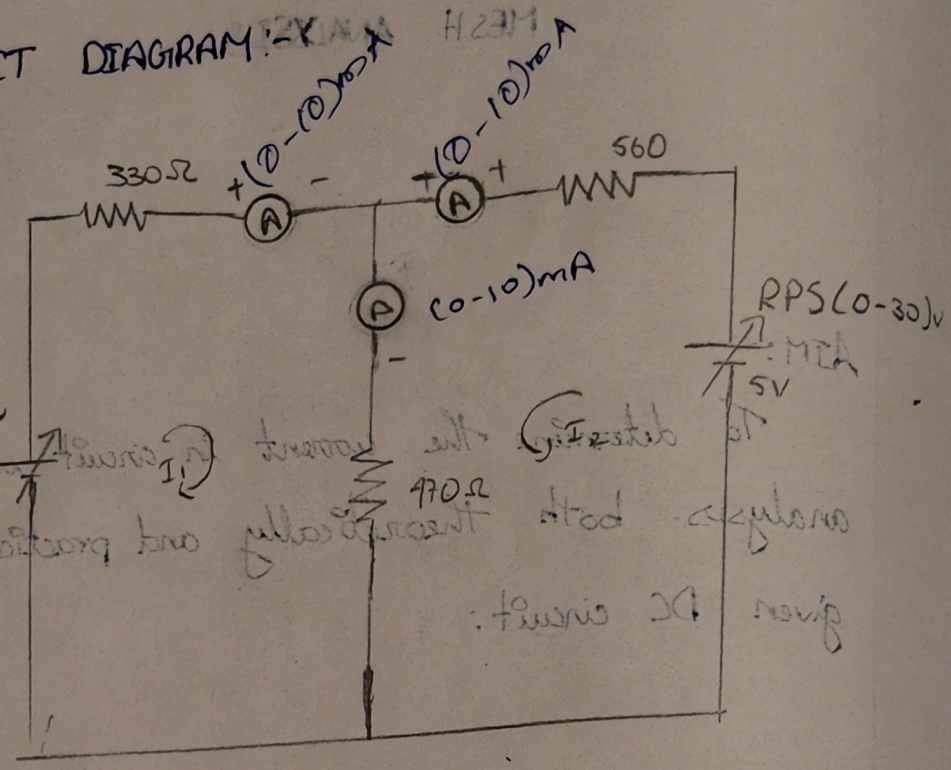
AIM:

To determine the current in circuit using mesh analysis. both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED:

SL.NO	APPARATUS	SPECIFICATION	QUANITY
1.	Regulated Power supply (RPS)	(0-30 V)	1
2.	Ammeter	(0-10mA) MC	3
3.	Resistor	330Ω, 470Ω, 560Ω	Each 1
4.	Bread board	—	1

CIRCUIT DIAGRAM:-



TABULAR COLUMNS:-

Parameters	Theoretical	Practical.
I_1	1.89 A	1.22 amphre
I_2	+4.21 A	-4.29 amphre
$I_3 = I_1 + I_2$	~~3.05A~~ 5.51 A	~~3.87~~ 5.44 amphre

THEORY:

Multi source DC circuits may be analyzed using a mesh current technique. The process involves identifying minimum number of small loops such that every component exists in at least one loop. KVL is then applied to each loop. meshes with the surrounding loop current. As a result there will be a set of ~~simulation~~ simultaneous equation created, an unknown mesh current for each loop. Once the mesh currents are determined, various branch currents and component voltage may be derived.

PROCEDURE:

1. Give connections as per the circuit diagram.
2. Switch ON the supply, vary the RPS (Regulated Power Supply) and set a particular input voltage.
3. Note down the readings of ammeter and voltmeter and tabulate them.
4. Vary the RPS for different input voltages and note down the readings of all the meters.
5. Reduce the RPS to its minimum value and switch off the supply
6. Using the tabulated values, verify Kirchoff's law practically, and verify it theoretically.

CALCULATIONS.

loop 1

$$330 I_1 + 470 (I_1 - I_2) = 3$$

$$330 I_1 + 470 I_1 - 470 I_2 = 3$$

$$800 I_1 - 470 I_2 = 3 \rightarrow (1)$$

loop 2:-

$$560 I_2 + 470 (I_1 + I_2) = 5$$

$$560 I_2 + 470 I_1 + 470 I_2 = 5$$

$$-470 I_1 + 1030 I_2 = -5 \rightarrow (2)$$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 800 & -470 \\ -470 & 1030 \end{vmatrix} = 824000 - 220900$$

$$\Delta = 603,100$$

$$\Delta I_1 = \begin{vmatrix} 3 & -470 \\ -5 & 1030 \end{vmatrix} = 3090 - 2350$$

$$= 740$$

$$\frac{\Delta I_1}{\Delta} = 1.2 \text{ A}$$

$$\frac{\Delta I_2}{\Delta} = \begin{vmatrix} 800 & 3 \\ -470 & -5 \end{vmatrix} = -4000 + 1,410$$

$$= -2,590$$

$$\frac{\Delta I_2}{\Delta} = +4.2 \text{ A}$$

$$I = I_1 + I_2 = 5.4$$

Sl.No	Description	Marks Alloted	Mark obtained.
1.	Experimental Setup/Connection	20 mark	18
2.	Observation/reading	20 mark	18
3.	Calculation /Result	20 mark	18
4.	Viva	20 mark	18
5	Record	20 mark	18
		Total 100	90

Result:-

Thus the Mesh analysis is verified practically and theoretically.

The resultant current for 3 / 5 V supply are:

a) The current across 330Ω resistor is 1.2 A.

b) The current across 560Ω resistor is 4.2 A.

c) The current across 470Ω resistor is 5.2 A.